ARUL RAJ R

Aeronautical Engineering Student
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PROFESSIONAL SUMMARY

Dedicated Aeronautical Engineering student pursuing a B.E at Excel Engineering College (CGPA 8/10). Strong theoretical knowledge of aerodynamics, aircraft structures, propulsion systems, and UAV technologies, complemented by hands-on experience through three industrial internships and team-led RC aircraft and drone projects.

EDUCATION

B.E – Aeronautical Engineering **2023 – 2027**
Excel Engineering College | CGPA: 8/10

TECHNICAL SKILLS

Design & CAD: SolidWorks (Basics), CATIA (Basics), Autodesk Fusion 360
Analysis & Simulation: ANSYS Mechanical APDL, ANSYS Workbench, Finite Element Analysis (FEA), Flight Simulator
Aerospace Skills: UAV Systems – Drone Assembly, Flight Controller Calibration, Aircraft Design & Development, Glider / RC Aircraft Fabrication
Software: Microsoft Word, Microsoft Office, Excel, PowerPoint
Languages: Tamil, English

INTERNSHIP EXPERIENCE

Industrial Intern | ZF WABCO India Pvt. Ltd., Ambattur **Jun - Sept 2026**
- Worked in the Actuator Assembly Department and gained practical exposure to commercial vehicle brake actuator manufacturing.
- Performed final inspection, rework, and retesting operations to ensure product quality.
- Learned industrial manufacturing processes, quality control procedures, and production line operations.
- Understood lean manufacturing concepts, safety standards, and production workflow.

Aerospace Intern | Aakash Aerospace Pvt. Ltd. **Jan 2026**
- Assisted in aircraft component design and engineering analysis.
- Learned UAV manufacturing processes and aerospace production techniques.

Engineering Intern | Coral Coil Manufacturing Pvt. Ltd. **Jan 2025**
- Observed industrial coil manufacturing and quality inspection processes.
- Gained practical knowledge of manufacturing operations and production workflow.

TECHNICAL PROJECTS

Design & Fabrication of RC Aircraft Using EPP Foam **2026**
- Led a team of four members in designing and fabricating a functional RC aircraft.
- Performed aerodynamic calculations including lift-to-drag ratio, wing loading, and stall speed.
- Integrated electronic components and successfully completed aircraft assembly.

Aerodynamic Cycle Analysis | Aircraft Propulsion Research Project **2026**
- Studied the complete aerodynamic cycle used in aircraft propulsion systems.
- Analyzed compressor, combustion chamber, turbine, and nozzle performance.
- Evaluated thrust generation, thermal efficiency, and pressure variations.

Structural Analysis of Piston | ANSYS Workbench **2025**
- Performed finite element analysis to determine Von-Mises stress and total deformation.
- Validated structural strength through simulation and engineering calculations.

Drone Healthcare Delivery System **2025**
- Designed a conceptual UAV system for emergency medical supply transportation.
- Studied payload capacity, autonomous navigation, and route optimization techniques.

CERTIFICATIONS

- ANSYS Simulation Innovation Workshop – NIT Trichy Synergy 2026
- Essential Skills for CAD Design Engineers – C CUBE Technologies, Erode
- SolidWorks Essentials Workshop – Excel Engineering College
- Drone Technology Workshop – Excel Engineering College
- MECHFORGE 2K’25 – Vel Tech High Tech College
- ETABS – Software Training – PSG Institute of Technology and Applied Research